

智能阶梯轴自动上料与AI视觉检测系统项目书

60x20 mm工件 | 0.12 kg | 12件/分钟

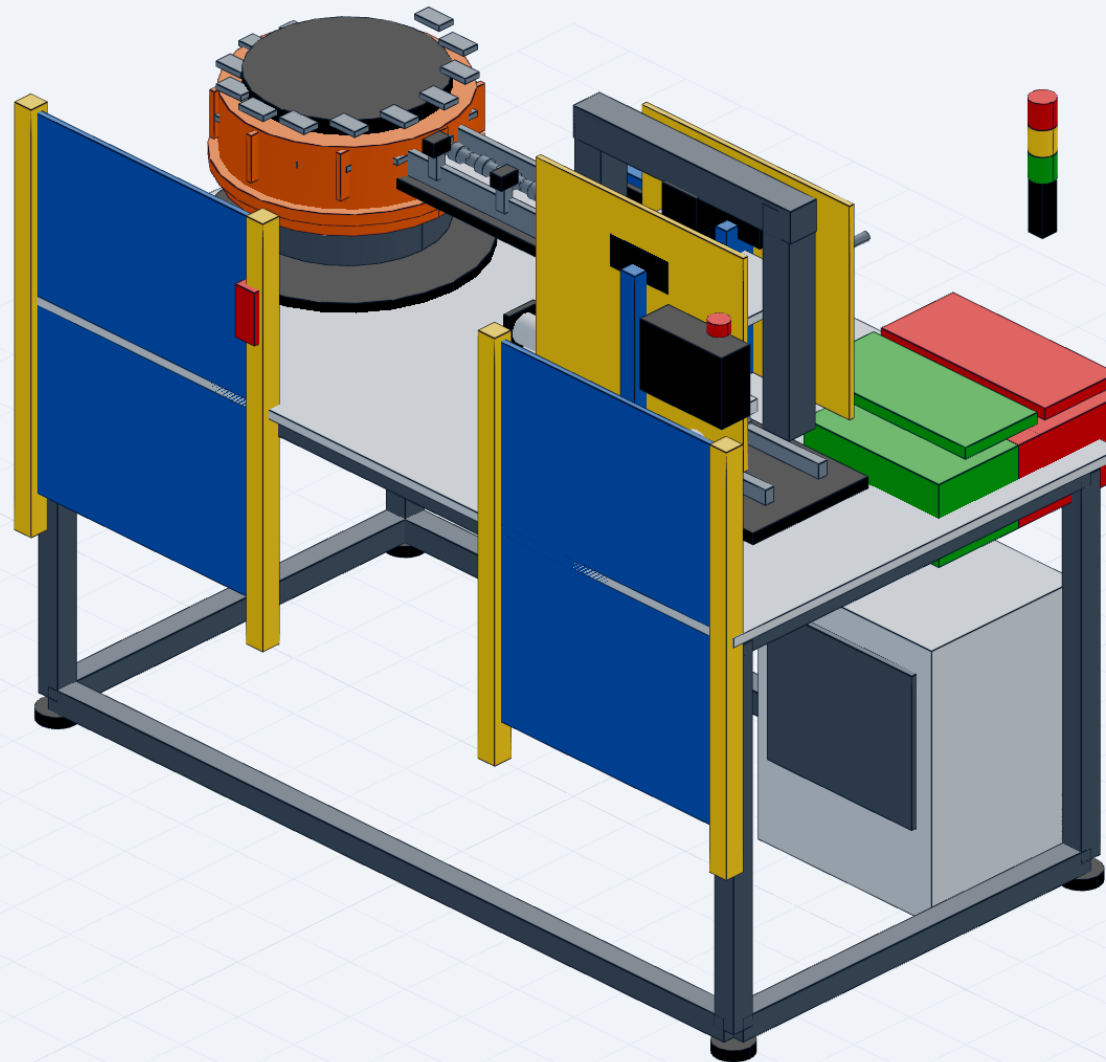
振动盘定向、直线送料、双气缸分料、V形定位、视觉/激光检测和OK/NG分流完整工作站。

离散螺旋轨道与可调直线送料
5 s循环动作分解
视觉轮廓与激光测径组合
阀岛、电控、安全门和防护一体化

Editable CAD source + STEP + GLB + calculations + BOM + manufacturing plan + validation evidence

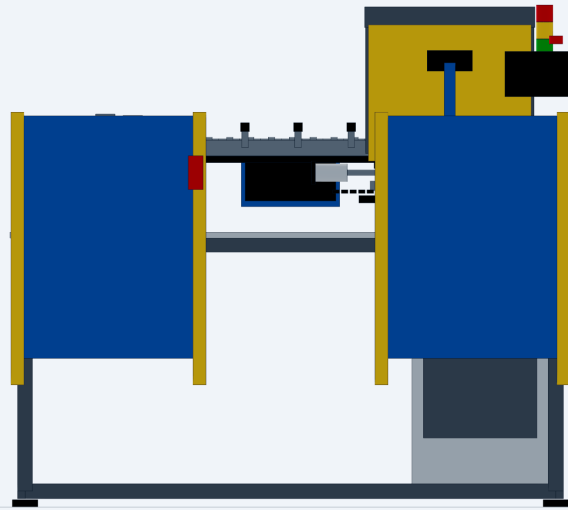
01 Detailed assembly overview

ISO

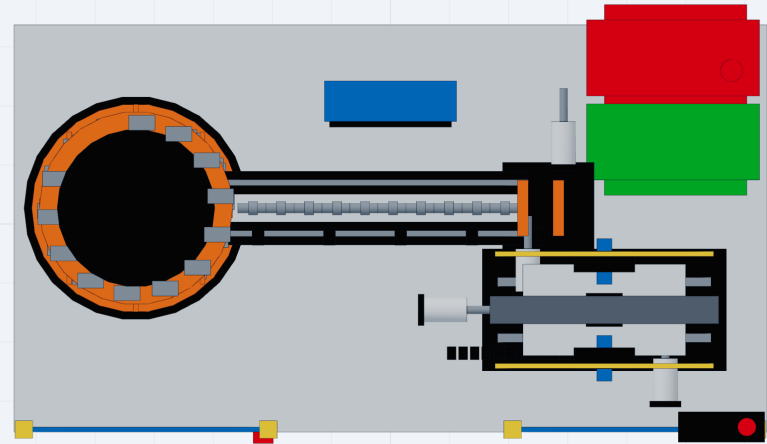


02 Orthographic and opposite views

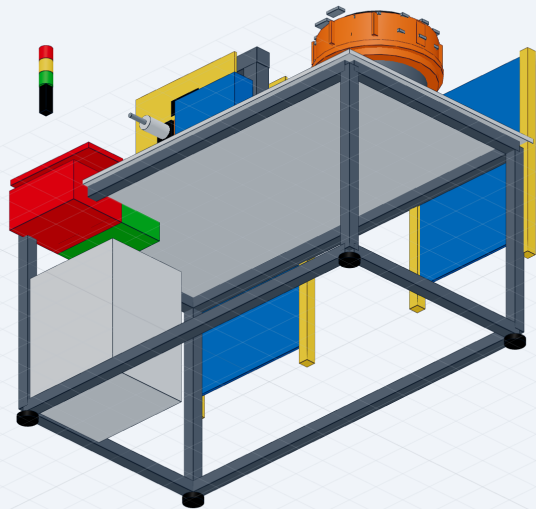
FRONT



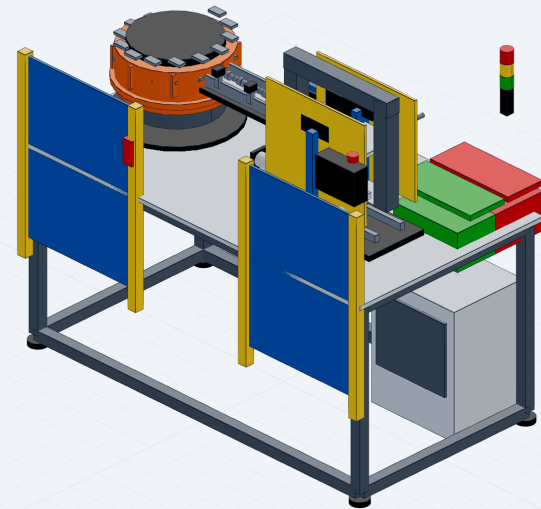
TOP



ISO



ISO



03 Parameters and preliminary calculations

parameter	value	unit	status
workpiece_length	60	mm	design baseline
workpiece_max_diameter	20	mm	design baseline
workpiece_mass	0.12	kg	design baseline
target_rate	12	parts/min	target
cycle_time	5	s	calculated
air_pressure	0.5	MPa	assumed
transfer_cylinder_bore	20	mm	selected baseline
separator_cylinder_bore	16	mm	selected baseline
model_bounding_box	1480x862.5x1324.5	mm	verified
step_faces	924	count	verified
step_edges	1698	count	verified

One 5 s cycle is allocated as: confirmation 0.4 s, separation 0.7 s, transfer 1.0 s, settle 0.4 s, image/laser acquisition 1.0 s, judgement 0.3 s, eject/return 1.2 s.

For a 0.12 kg part on a guide with assumed friction coefficient 0.2 and 0.5 m/s² acceleration, required force with factor 3 is below 1 N. A 16 mm bore cylinder at 0.5 MPa provides about 100 N theoretical force; a 20 mm transfer cylinder provides about 157 N. This leaves margin for seals, guide friction and occasional misalignment.

Five photoelectric checks cover bowl exit, track queue, first gate, second gate and V-block arrival. Laser micrometer measures the critical diameter; the top camera checks overall contour, step position, orientation and surface-defect candidates.

04 BOM, manufacturing and validation

item	component	qty	specification
1	vibratory bowl feeder	1	400 mm class
2	linear feeder track	1	620 mm adjustable rails
3	separator cylinders	2	16 mm bore / 65 mm stroke
4	transfer cylinder	1	20 mm bore / 170 mm stroke
5	eject cylinder	1	16 mm bore / 85 mm stroke
6	linear guides	2	420 mm
7	V-block fixture	1	hardened replaceable inserts
8	photoelectric sensors	5	24 VDC
9	industrial camera	1	top view
10	telecentric lens	1	matched field of view
11	backlight	1	320 x 180 mm
12	laser micrometer pair	1	emitter/receiver
13	pneumatic valve island	1	8-station
14	PLC/HMI/electrical cabinet	1 set	24 VDC control
15	OK and NG bins	2	removable
16	aluminium-profile frame	1 set	40 series
17	polycarbonate guarding	1 set	door interlock
18	cable chain	1	transfer axis

Manufacturing / assembly

Track and fixture surfaces are CNC milled with replaceable wear strips. V-block inserts are tool steel, heat treated and ground. Frame members are cut aluminium profiles with dowel-located station plates. Sensor brackets use slots for calibration. Guard panels are polycarbonate in aluminium frames. Pneumatic tubing and electrical cables are separated and labelled for maintenance.

Validation boundary